Indian Institute of Technology Kanpur

Department of Space, Planetary & Astronomical Sciences & Engineering (SPASE)

SPA617: Radio Astronomy Laboratory

RF Transmission, Reflections, and Standing Waves

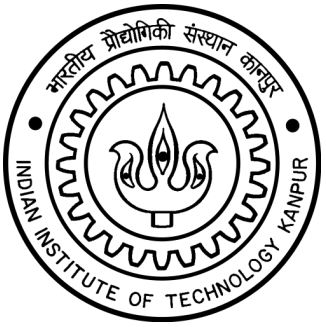
Laboratory Report

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Abstract

This experiment investigates signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies. The experiment is used to determine important cable parameters experimentally, including the phase velocity, velocity factor, reflection coefficient, characteristic impedance, inductance and capacitance per unit length, and cable attenuation.

1 Introduction and Objectives

A coaxial transmission line is used to guide electromagnetic energy from one point to another. In this experiment, a coaxial cable is studied at radio frequencies to understand signal propagation, reflection, and the formation of standing-wave behaviour. When a travelling wave encounters a load whose impedance differs from the characteristic impedance of the cable, part of the wave is reflected towards the source. The incident and reflected waves then superimpose, producing constructive and destructive interference at the observation point.

1.1 Our Aim & Objectives

The aim of this experiment is to understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

Key objectives of the experiment are:

1. To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.
2. To study signal reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load.
3. To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a $50\,\Omega$ termination, indicating that its characteristic impedance is close to $50\,\Omega$.
4. To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\,\Omega$.
5. To measure the insertion loss and attenuation per unit length of the coaxial cable.

2 Components & instruments

The following components and instruments were used in the experiment:

1. RF signal generator
2. Digital storage oscilloscope (DSO)
3. Spectrum analyser
4. RF Tee connector

5. Coaxial cable under test
6. Short reference coaxial cable
7. 50 Ω termination
8. Open-circuit termination, obtained by leaving the cable end unconnected
9. Suitable RF adapters and connectors

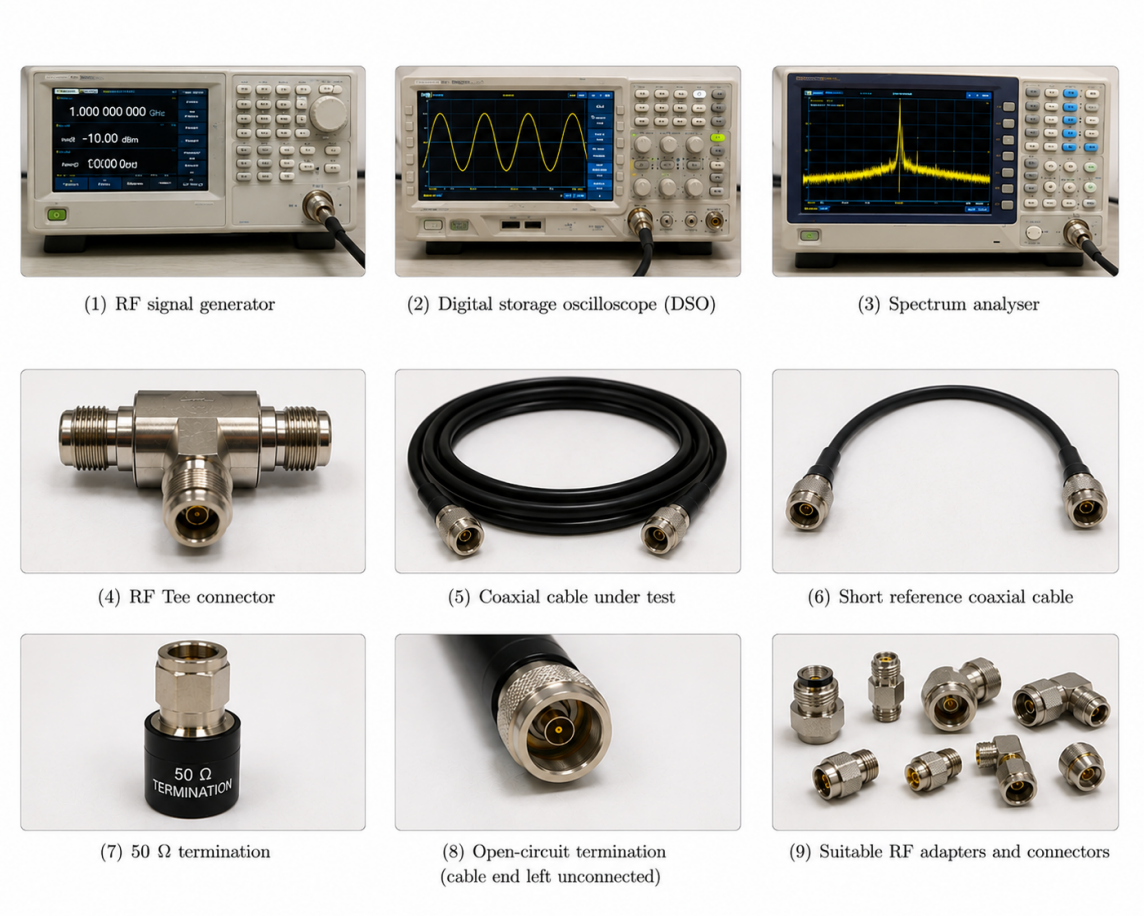


Figure 1: RF instruments and components used in the experiment.

3 Theory

A transmission line guides electromagnetic energy from one point to another; for example, a coaxial cable is a transmission line that is used to carry radio-frequency (RF) signals. It consists of an inner conductor, a dielectric material, and an outer conductor that acts as a shield. The cable guides the electromagnetic energy along its length, and its speed in the cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

An important property of a transmission line is its characteristic impedance, denoted by Z_0. So, when the travelling signal reaches the end of the cable, its behaviour depends on the load impedance Z_L and the characteristic impedance Z_0 of the cable. If the two impedances are different, a part of the signal is reflected back towards the source. The incident and reflected signals combine at the observation point. Depending on their phase relationship, they can

reinforce or partially cancel each other, producing larger or smaller measured voltages. Those voltages are measured using the oscilloscope at a fixed Tee junction.

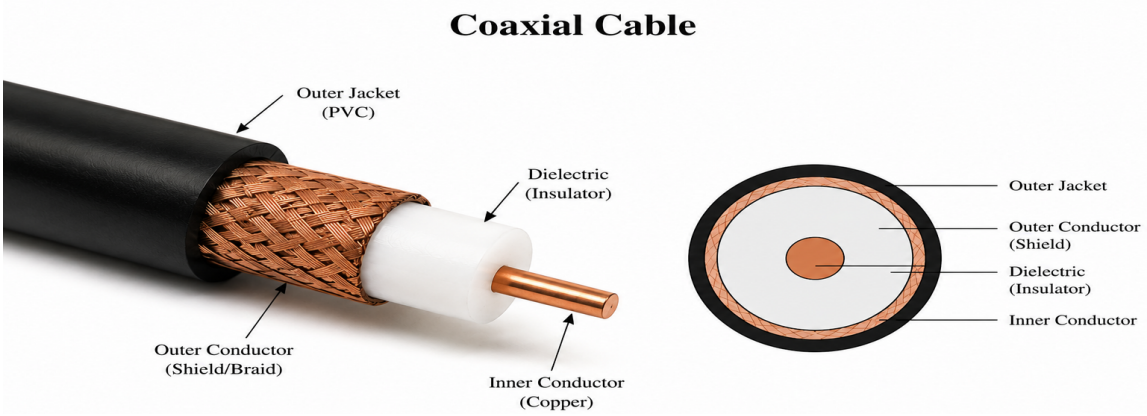


Figure 2: RF instruments and components used in the experiment.

3.1 Phase Velocity Measurement

Let the length of the coaxial cable be ℓ and the phase velocity of the signal in the cable be v_p. Since the reflected signal travels from the measurement point to the end of the cable and back, it covers a round-trip distance of 2ℓ. The corresponding round-trip time is therefore

$$\tau = \frac{2\ell}{v_p} \tag{1}$$

The phase of the reflected signal at the measurement point depends on the signal frequency and the round-trip delay. It can be written as

$$\phi_{\mathrm{RT}} = 2\pi f\tau + \phi_{\Gamma}, \tag{2}$$

where ϕ_{Γ} represents the phase shift introduced by the termination. For an ideal open circuit, this phase shift is taken to be zero.

When the frequency is changed from one maximum to the next maximum, or from one minimum to the next minimum, the round-trip phase changes by 2π. Therefore,

$$2\pi\Delta f\,\tau = 2\pi, \tag{3}$$

which gives

$$\tau = \frac{1}{\Delta f} \tag{4}$$

Combining this result with the expression for the round-trip time gives the phase velocity of the signal:

$$v_p = 2\ell\Delta f, \tag{5}$$

where Δf is the frequency separation between two consecutive maxima or two consecutive minima.

The velocity factor of the cable is then calculated by comparing the measured phase velocity with the speed of light in vacuum:

$$VF = \frac{v_p}{c}, \tag{6}$$

where

$$c \approx 3 \times 10^8 \text{ m/s} \tag{7}$$

3.2 Reflection Coefficient and Characteristic Impedance

The amount of signal reflected from a load is described by the voltage reflection coefficient,

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}, \tag{8}$$

where Z_L is the load impedance and Z_0 is the characteristic impedance of the coaxial cable. For an open circuit, $\Gamma = +1$, while for a short circuit, $\Gamma = -1$. For a perfectly matched load, $Z_L = Z_0$ and therefore $\Gamma = 0$.

At the Tee junction, the measured voltage is the combination of the incident and reflected signals. The maximum voltage occurs when the two signals interfere constructively,

$$V_{\max} = |V_{\text{incident}}| + |V_{\text{reflected}}|, \tag{9}$$

while the minimum occurs when they interfere destructively,

$$V_{\min} = |V_{\text{incident}}| - |V_{\text{reflected}}| \tag{10}$$

From these two measurements, the incident and reflected voltage amplitudes can be written as

$$|V_{\text{incident}}| = \frac{V_{\max} + V_{\min}}{2}, \tag{11}$$

and

$$|V_{\text{reflected}}| = \frac{V_{\max} - V_{\min}}{2} \tag{12}$$

Hence, the magnitude of the reflection coefficient can be estimated from

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{13}$$

In this experiment, the reflection from an open-ended cable is compared with the reflection obtained using a $50\,\Omega$ termination. A much smaller reflection with the $50\,\Omega$ termination indicates that the cable is approximately matched to this load, suggesting that its characteristic impedance is close to $50\,\Omega$.

3.3 Inductance and Capacitance per Unit Length

The coaxial cable can also be described by its inductance and capacitance per unit length. Let L' be the inductance per unit length in H/m and C' be the capacitance per unit length in F/m. For a low-loss transmission line,

$$v_p = \frac{1}{\sqrt{L'C'}} \tag{14}$$

and

$$Z_0 = \sqrt{\frac{L'}{C'}} \tag{15}$$

Here, the characteristic impedance of the cable can also be calculated as

$$Z_0 = Z_L \frac{1 - |\Gamma|}{1 + |\Gamma|} \tag{16}$$

$$\tag{17}$$

Using the measured phase velocity and the approximate characteristic impedance of the cable, these parameters are calculated as

$$L' = \frac{Z_0}{v_p} \tag{18}$$

and

$$C' = \frac{1}{Z_0 v_p} \tag{19}$$

3.4 Cable Attenuation

Cable attenuation describes the reduction in signal level as the RF signal travels through the coaxial cable. In this experiment, the signal power is measured in dBm using a spectrum analyser.

First, a reference power P_{ref} is measured using the short reference connection. The cable under test is then inserted and the power P_{cable} is measured again. The insertion loss is therefore

$$IL = P_{\text{ref}} - P_{\text{cable}} \tag{20}$$

The attenuation per unit length is obtained by dividing the insertion loss by the physical length ℓ of the cable:

$$\alpha = \frac{IL}{\ell} \text{ dB/m} \tag{21}$$

4 Experimental Setup

For the phase-velocity and reflection measurements, the RF signal generator, RF Tee connector, coaxial cable under test, and digital storage oscilloscope (DSO) were connected as shown in Fig. 3. The signal generator was connected to the Tee connector, with one port of the Tee connected to the DSO and the other connected to the coaxial cable under test. The far end of the cable was left open for the open-circuit measurement and connected to a 50 Ω termination for the matching measurement.

And for the cable attenuation measurement, the oscilloscope was replaced by a spectrum analyser. The RF signal generator was connected to the spectrum analyser using a short reference cable and the required RF adapters as shown in Fig. 4. The reference connection was first used to measure P_{ref}. It was then replaced by the coaxial cable under test, while keeping the remaining connections unchanged, to measure P_{cable}.

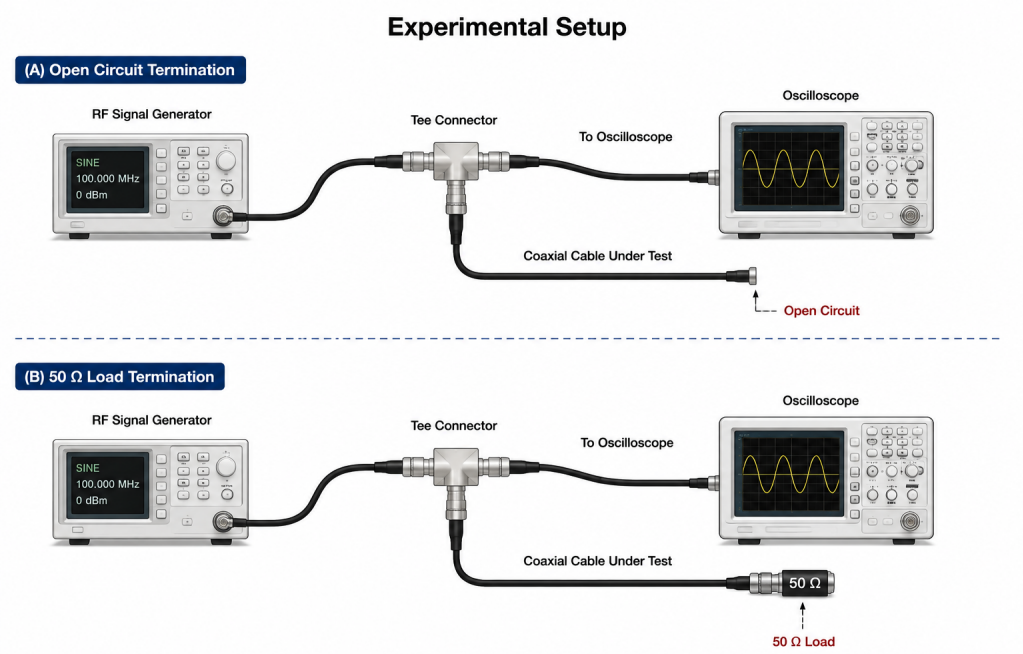


Figure 3: Experimental setup for the open-circuit and 50 Ω termination measurements.

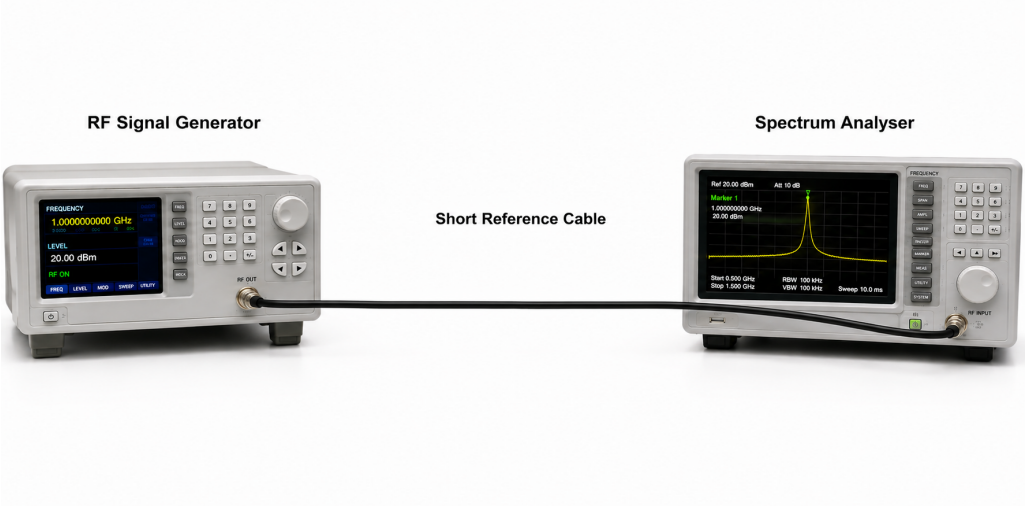


Figure 4: Setup used for measuring the cable attenuation with the spectrum analyser.

5 Experimental Procedure

We completed the experiment in the following parts. Before starting the measurements, the physical length of the coaxial cable was measured. The RF signal generator, Tee connector, coaxial cable under test, and the required measuring instrument were then connected according to the experimental setup.

5.1 Part A and Part B: Open-Circuit and 50 Ω Termination

For Parts A and B, the RF signal generator, Tee connector, coaxial cable, and DSO were connected together [See Fig. 3]. The DSO input impedance was set to 1 MΩ, and the signal-

generator amplitude was kept constant.

Initially, the far end of the cable was left open. The frequency was swept over the required range, and the voltage at the Tee junction was recorded at suitable frequency intervals. The measured data were used to identify consecutive maxima or minima and determine the frequency spacing for the phase-velocity and velocity-factor calculations.

For the reflection measurement, the open-circuit maximum and minimum voltages were used to calculate $|\Gamma_{\text{open}}|$. The open termination was then replaced by a $50\,\Omega$ load, and the frequency was swept over the same range without changing the signal-generator amplitude. The corresponding maximum and minimum voltages were recorded to determine $|\Gamma_{50}|$. The two reflection coefficients were then compared.

5.2 Part C: Inductance and Capacitance per Unit Length

The measured phase velocity and the approximate characteristic impedance $Z_0 \approx 50\,\Omega$ were used to calculate the inductance and capacitance per unit length of the coaxial cable.

5.3 Part D: Cable Attenuation

For Part D, the DSO and Tee-junction setup were replaced by a direct connection between the RF signal generator and spectrum analyser. A short reference cable was first connected [See Fig. 4], and the reference power P_{ref} was recorded at the selected frequency and input power.

The short reference cable was then replaced by the cable under test without changing the remaining setup. The measured power P_{cable} was recorded. The insertion loss and attenuation per unit length were calculated from these measurements and the cable length.

6 Observations and Calculations

6.1 Cable and Setup Parameters

Parameter	Value
Physical cable length, ℓ	0.92 m
DSO input impedance	$1\,\text{M}\Omega$

Table 1: Fixed physical and setup parameters used in the experiment.

6.2 Part A: Phase Velocity and Velocity Factor

The voltage at the Tee junction was measured while sweeping the frequency of the RF signal with the far end of the coaxial cable left open.

Table 2: Measured voltage as a function of frequency for the open-ended coaxial cable.

Frequency (MHz)	Voltage (V)	Observation
57	0.060	Minimum
62	0.225	
68	0.578	
70	0.693	
75	0.974	
80	1.280	
85	1.510	

Frequency (MHz)	Voltage (V)	Observation
90	1.610	
95	1.710	
97	1.720	Maximum
100	1.690	
105	1.650	
110	1.650	
115	1.580	
120	1.540	
125	1.500	
130	1.420	
135	1.330	
140	1.230	
145	1.080	
150	0.932	
155	0.762	
160	0.579	
165	0.395	
170	0.200	
171	0.163	
172	0.121	
173	0.090	
174	0.080	Minimum
175	0.097	

From the measured data, the prominent maximum occurs at approximately 97 MHz with a voltage of 1.720 V. The measured minima occur at approximately 57 MHz and 174 MHz, with voltages of 0.060 V and 0.080 V, respectively.

Calculation

The physical length of the coaxial cable was measured as

$$\ell = 92\,\text{cm} = 0.92\,\text{m} \tag{22}$$

From the measured voltage-frequency data, two minima were identified at

$$f_1 = 57\,\text{MHz} \tag{23}$$

and

$$f_2 = 174\,\text{MHz} \tag{24}$$

The frequency spacing between the two consecutive minima is

$$\Delta f = |f_2 - f_1| \tag{25}$$

Therefore,

$$\Delta f = |174 - 57|\,\text{MHz} \tag{26}$$
$$= 117\,\text{MHz} \tag{27}$$
$$= 117 \times 10^6\,\text{Hz} \tag{28}$$

The phase velocity of the signal in the coaxial cable is given by

$$v_p = 2\ell\Delta f \tag{29}$$

Substituting $\ell = 0.92\,\text{m}$ and $\Delta f = 117 \times 10^6\,\text{Hz}$,

$$v_p = 2(0.92)(117 \times 10^6) \tag{30}$$
$$= 2.1528 \times 10^8\,\text{m/s} \tag{31}$$

Therefore, the phase velocity is

$$\boxed{v_p \approx 2.15 \times 10^8\,\text{m/s}} \tag{32}$$

The velocity factor is calculated using

$$VF = \frac{v_p}{c}, \tag{33}$$

where

$$c = 3 \times 10^8\,\text{m/s} \tag{34}$$

Hence,

$$VF = \frac{2.1528 \times 10^8}{3 \times 10^8} \tag{35}$$
$$\approx 0.718 \tag{36}$$

Thus, the velocity factor of the coaxial cable is approximately

$$\boxed{VF \approx 0.72} \tag{37}$$

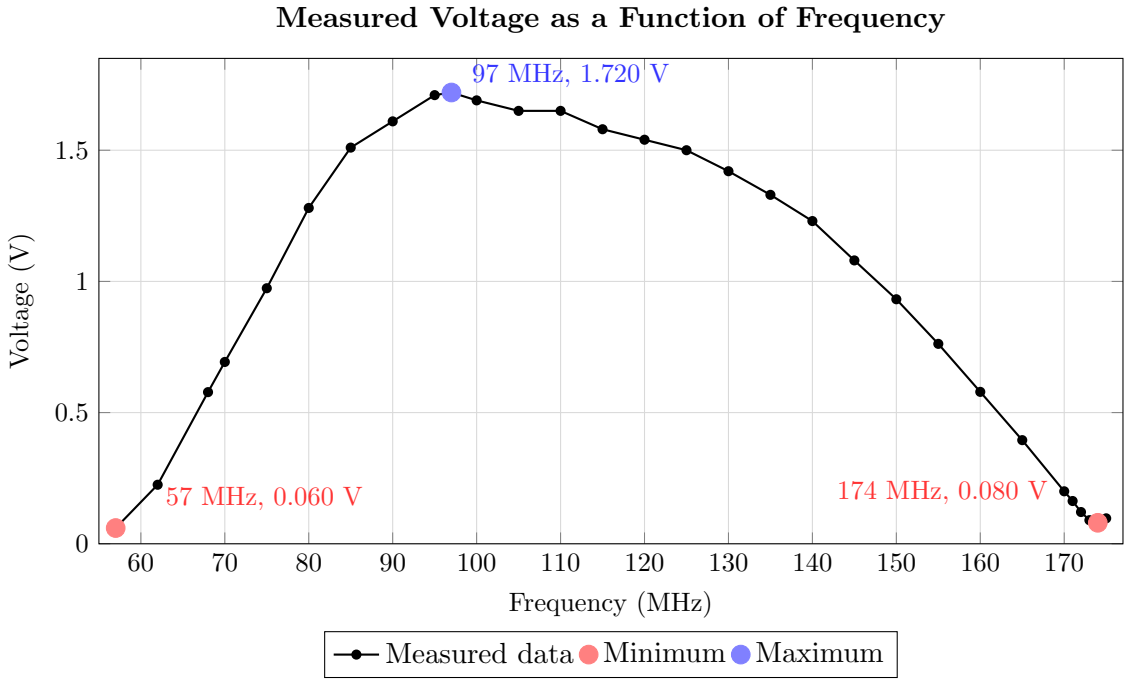


Figure 5: Measured voltage as a function of frequency for the open-ended coaxial cable. The identified maxima and minima are highlighted.

6.3 Part B: Reflection Coefficient and Characteristic Impedance

The maximum and minimum voltage amplitudes were obtained for both the open-circuit and $50\,\Omega$ termination cases. The magnitude of the reflection coefficient was calculated using

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{38}$$

Calculation

For the open-circuit termination,

$$V_{\max,\text{open}} = 1.720\,\text{V}, \qquad V_{\min,\text{open}} = 0.080\,\text{V} \tag{39}$$

Therefore,

$$|\Gamma_{\text{open}}| = \frac{V_{\max,\text{open}} - V_{\min,\text{open}}}{V_{\max,\text{open}} + V_{\min,\text{open}}} \tag{40}$$

$$= \frac{1.720 - 0.080}{1.720 + 0.080} \tag{41}$$

$$= 0.911 \tag{42}$$

Thus,

$$\boxed{|\Gamma_{\text{open}}| \approx 0.911} \tag{43}$$

For the $50\,\Omega$ termination,

$$V_{\max,50} = 1.090\,\text{V}, \qquad V_{\min,50} = 0.881\,\text{V} \tag{44}$$

Therefore,

$$|\Gamma_{50}| = \frac{V_{\max,50} - V_{\min,50}}{V_{\max,50} + V_{\min,50}} \tag{45}$$

$$= \frac{1.090 - 0.881}{1.090 + 0.881} \tag{46}$$

$$\approx 0.106. \tag{47}$$

Thus,

$$\boxed{|\Gamma_{50}| \approx 0.106} \tag{48}$$

The measured values and calculated reflection coefficients are summarized in Table 3.

Termination	$V_{\max}$ **(V)**	$V_{\min}$ **(V)**	$\lvert\Gamma\rvert$
Open circuit	1.720	0.080	0.911
$50\,\Omega$ load	1.090	0.881	0.106

Table 3: Measured maximum and minimum voltages and calculated reflection coefficients for the two terminations.

The reflection coefficient decreases substantially when the open termination is replaced by the 50 Ω load, from approximately 0.911 to 0.106. This indicates a much smaller reflected signal for the loaded case. The measured reflection coefficient for the 50 Ω termination was

$$|\Gamma_{50}| = 0.106$$

Using this value and the known load impedance $Z_L = 50\,\Omega$, the characteristic impedance was estimated using

$$\begin{aligned} Z_0 &= Z_L \frac{1 - |\Gamma_{50}|}{1 + |\Gamma_{50}|} && (49)\\ &= 50 \frac{1 - 0.106}{1 + 0.106} && (50)\\ &= 50 \frac{0.894}{1.106} && (51)\\ &\approx 40.4\,\Omega && (52) \end{aligned}$$

Therefore, the approximate characteristic impedance of the cable was found to be

$$\boxed{Z_0 \approx 40.4\,\Omega} \tag{53}$$

The substantial reduction in the reflected signal indicates that the 50 Ω termination provides a good match to the cable.

6.4 Part C: Inductance and Capacitance per Unit Length

Using the phase velocity obtained in Part A and the approximate characteristic impedance obtained from the reflection measurement in Part B,

$$v_p = 2.1528 \times 10^8\,\text{m/s} \tag{54}$$

and

$$Z_0 \approx 40.4\,\Omega. \tag{55}$$

The inductance and capacitance per unit length are calculated using

$$L' = \frac{Z_0}{v_p} \tag{56}$$

and

$$C' = \frac{1}{Z_0 v_p}. \tag{57}$$

The inductance per unit length is

$$\begin{aligned} L' &= \frac{40.4}{2.1528 \times 10^8} && (58)\\ &\approx 1.877 \times 10^{-7}\,\text{H/m} && (59) \end{aligned}$$

Therefore,

$$\boxed{L' \approx 1.88 \times 10^{-7}\,\text{H/m}} \tag{60}$$

or

$$\boxed{L' \approx 188\,\text{nH/m}} \tag{61}$$

The capacitance per unit length is

$$C' = \frac{1}{40.4(2.1528 \times 10^8)} \tag{62}$$

$$\approx 1.150 \times 10^{-10}\,\text{F/m} \tag{63}$$

Therefore,

$$\boxed{C' \approx 1.15 \times 10^{-10}\,\text{F/m}} \tag{64}$$

or

$$\boxed{C' \approx 115\,\text{pF/m}} \tag{65}$$

Quantity	**Calculated value**
Phase velocity, v_p	$2.1528 \times 10^8\,\text{m/s}$
Characteristic impedance, Z_0	$40.4\,\Omega$
Inductance per unit length, L'	$188\,\text{nH/m}$
Capacitance per unit length, C'	$115\,\text{pF/m}$

Table 4: Calculated transmission-line parameters.

6.5 Part D: Cable Attenuation

For the attenuation measurement, the reference power and the power measured after inserting the cable under test were recorded at several frequencies. The cable length used for the calculation was

$$\ell = 0.92\,\text{m}.$$

The insertion loss and attenuation per unit length were calculated using Equations (20) and (21). The resulting values are summarized in Table 5.

Frequency (GHz)	P_{ref} **(dBm)**	P_{cable} **(dBm)**	$IL = P_{\text{ref}} - P_{\text{cable}}$ **(dB)**	$\alpha = IL/\ell$ **(dB/m)**
0.5	20.00	18.50	1.50	1.63
1.0	20.00	17.45	2.55	2.77
1.6	20.00	17.00	3.00	3.26
2.0	20.00	16.49	3.51	3.82

Table 5: Measured power, insertion loss, and attenuation of the coaxial cable.

The results show that the insertion loss increases with frequency. As a result, the attenuation per unit length also increases, from 1.63 dB/m at 0.5 GHz to 3.82 dB/m at 2.0 GHz.

7 Final Results

The main measured and calculated results obtained from the experiment are summarized in Table 6.

Quantity	**Symbol**	**Result**
Cable length	ℓ	0.92 m
Frequency spacing	Δf	117 MHz
Phase velocity	v_p	2.15×10^8 m/s
Velocity factor	VF	0.72
Open-circuit reflection coefficient	$\lvert\Gamma_{\text{open}}\rvert$	0.911
50 Ω reflection coefficient	$\lvert\Gamma_{50}\rvert$	0.106
Characteristic impedance	Z_0	40.4 Ω
Inductance per unit length	L'	188 nH/m
Capacitance per unit length	C'	115 pF/m
Attenuation at 0.5 GHz	α	1.63 dB/m
Attenuation at 1.0 GHz	α	2.77 dB/m
Attenuation at 1.6 GHz	α	3.26 dB/m
Attenuation at 2.0 GHz	α	3.82 dB/m

Table 6: Summary of the measured and calculated results.

8 Conclusion

The experiment demonstrated the propagation, reflection, standing-wave behaviour, and attenuation of an RF signal in a coaxial transmission line. The voltage measured at the Tee junction varied periodically with frequency because of the interference between the incident and reflected signals.

From the frequency spacing between consecutive minima, the phase velocity of the cable was determined to be approximately

$$v_p \approx 2.15 \times 10^8 \,\text{m/s},$$

giving a velocity factor of

$$VF \approx 0.72.$$

The reflection measurements showed a large reflection for the open-ended cable, with

$$|\Gamma_{\text{open}}| = 0.911.$$

When the cable was terminated with a 50 Ω load, the reflection coefficient decreased to

$$|\Gamma_{50}| = 0.106.$$

This large reduction in reflection shows that the 50 Ω termination provides a good match to the cable. The characteristic impedance estimated from the measured reflection was approximately 40.4 Ω. The difference from the nominal 50 Ω value may be due to measurement uncertainties, connector and adapter effects, Tee-junction losses, termination imperfections, and uncertainty in locating the exact voltage maxima and minima.

Using the measured phase velocity and the approximate value $Z_0 \approx 40.4\,\Omega$, the inductance and capacitance per unit length were calculated as

$$L' \approx 188\,\text{nH/m}$$

and

$$C' \approx 115\,\text{pF/m},$$

respectively.

The attenuation measurements showed that the cable loss increased with frequency. The attenuation increased from approximately 1.63 dB/m at 0.5 GHz to 3.82 dB/m at 2.0 GHz.

Overall, the experiment confirmed the main transmission-line behaviour: impedance mismatch produces reflection and standing-wave variations, while a termination close to the characteristic impedance reduces the reflected signal.

Possible sources of measurement uncertainty:

The main sources of uncertainty include the finite frequency resolution when locating the maxima and minima, uncertainty in reading the voltage from the DSO, uncertainty in measuring the cable length, connector and adapter effects, variations in the signal-generator amplitude, and measurement uncertainty of the spectrum analyser. These factors can affect the measured reflection coefficient, phase velocity, characteristic impedance, and cable attenuation.

Use of Artificial Intelligence

Artificial intelligence tools were used as a learning and writing support during the preparation of this laboratory report. The main uses were:

- To understand the experimental setup, transmission-line concepts, reflections, standing waves, phase velocity, and reflection coefficient.
- To clarify the derivation and physical meaning of the equations used in the experiment, particularly the relation between frequency spacing, round-trip phase, and phase velocity.
- To organize the experimental procedure and present the laboratory measurements in a clear report structure.
- To assist with LaTeX and Overleaf formatting, including tables, equations, figures, captions, references, and document layout.
- To check calculations and help interpret the measured results, including the reflection coefficient, characteristic impedance, inductance, capacitance, insertion loss, and attenuation.
- To identify possible sources of experimental uncertainty and discuss differences between measured and expected values.
- To create simple illustrative diagrams for the experimental setup and electromagnetic-wave concepts.

The experimental measurements and observations were obtained during the laboratory work. The AI tool was used primarily to support understanding, calculation, presentation, and interpretation of the experiment.